

# Common bacterial Skin Infections

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## Common Bacterial Skin Infections

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### Introduction

Bacteria exist in the external environment, but the **intact normal skin has a strong resistance** to bacterial multiplication and invasion.

### Factors that enhance skin resistance

1. Relative **dryness** of normal skin
2. Low skin **pH of 5.5**
3. **Unsaturated long chain fatty acids** in skin lipids
4. Natural **antibacterial substances in the sebum**
5. Circulating **immunoglobulins**
6. **Cellular immunity**
7. **Delayed hypersensitivity**
8. **Bacterial interference** – the suppression of bacterial species or strains by other bacteria

### Factors that favour bacterial skin infections

1. **Trauma** – abrasions, scratches, cuts
2. **Insect bites**
3. **Increased humidity**
4. **Malnutrition**
5. **Severe illness**
6. **Low standards of personal or community hygiene**
7. **Coexisting skin disorders** such as atopic dermatitis
8. **Natural or acquired immune depression**
9. **Iatrogenic immune suppression**

### The common bacterial skin infections

- Folliculitis
- Furunculosis
- Carbuncles
- Impetigo
- Cellulitis
- Staphylococcal scalded skin syndrome

### Folliculitis

Folliculitis is the **superficial inflammation and infection of a single hair follicle**, appearing as small, itchy, pus-filled bumps. It affects skin around a hair follicle, usually causing tender, pus-filled inflammations.

### Aetiology

- Bacterial
- Fungal
- Viral
- Parasitic

### Bacterial folliculitis

Presents as **small, tender, red, or pus-filled bumps clustered around hair follicles**. Each bump resembles a small pimple with a **hair sticking out of the centre**, surrounded by inflamed, red, and warm skin.

**Sites:** most commonly the face, scalp, thighs, or buttocks.

## Treatment

- Warm compresses
- Antibacterial washes (like benzoyl peroxide)
- Topical antibiotics (e.g. **mupirocin** or **clindamycin**)
- Severe or stubborn cases may require oral antibiotics (like **cephalexin** or **doxycycline**)
- Avoid shaving, picking the bumps, and tight clothing, to prevent spreading
- Good personal hygiene

## Furunculosis (boils)

Furunculosis is the condition of having **multiple, deeper, and more severe infections of the hair follicles** known as boils (furuncles), which **extend into subcutaneous tissue**.

The causative agent is usually *S. aureus*, which penetrates abraded perifollicular skin.

The lesions may originate from a **preceding folliculitis**, or may arise initially as a deep-seated, tender, erythematous, perifollicular nodule.

## Progression

**Indurated lesion -> central necrosis -> suppuration -> rupture and discharge of a central core of necrotic tissue -> destruction of the follicle**

Healing occurs with **scar formation**.

## Sites of predilection

Hair-bearing areas: **scalp, face, back of neck, axillae, buttocks, and groin**.

## Predisposing factors

1. Obesity
2. Hyperhidrosis
3. Maceration
4. Friction
5. Pre-existing dermatitis

## Treatment

- Antibiotics
- Personal hygiene
- Hot moist compress
- Incision and drainage

## Carbuncles

Caused by *Staphylococcus aureus*.

A carbuncle is an infection of a **group of contiguous follicles**, with **multiple drainage points**, accompanied by inflammatory changes in the surrounding connective tissue.

The lesions are **painful, tender, firm to hard, indurated lumps**.

It may be accompanied by **fever, leukocytosis, and bacteraemia**.

## Sites of predilection

Back of neck, shoulder, hip and thigh.

## Treatment

- Oral or parenteral antibiotics such as **cloxacillin, cephalexin**
- **Cephalosporin, clindamycin, or erythromycin** for penicillin-allergic patients
- In **MRSA**: **rifampicin, cotrimoxazole**
- Attention to personal hygiene

## Impetigo

A **contagious bacterial infection of the epidermis**.

Transmitted by **direct contact with infected persons or fomites**.

Can affect any skin region, but is most common on the **face around the nose, mouth and hand**.  
Commonly affects children aged **4-7 years**.

**Primary impetigo**: more common in children; infection is through **minor breaks in the skin**.

**Secondary impetigo**: occurs at any age, on secondary infection of trauma or wounds.

Superficial blisters rupture easily, releasing yellow exudate that dries and forms a **honey-coloured crust**.

**Diagnosis** is clinical, but can be confirmed by **gram stain or culture** of crust or fluid from bullae.

## Bullous impetigo

- **Always caused by *S. aureus* phage group 2**
- Develops on **intact skin**
- Vesicles and **flaccid transparent bullae** (large vesicles) containing clear yellow or slightly turbid fluid, with or without surrounding erythema
- Occurs more on **face, buttocks, trunk, perineum, and extremities**
- Common in **neonates and children under 5 years**
- **Systemic symptoms are common**

## Non-bullous impetigo

- **More common** than bullous
- Usually due to ***S. aureus* or *S. pyogenes* or mixed infection**
- Typically affects **trauma sites**
- Presents as scattered discrete **1-3 cm lesions with honey-coloured crust** and surrounding erythema
- Most common **around mouth/nose**
- Patients may have **lymphadenopathy**

**Predisposing lesions**: insect bites, abrasions, lacerations, chickenpox, scabies, pediculosis, and burns.

The infection may be spread to other parts of the body by the **fingers, clothing, and towels**.

## Treatment

- Local infections are treated with **topical saline or aluminium acetate**, then **2% mupirocin ointment**
- **Normal saline or potassium permanganate soaks** will help remove the crusts
- **Mupirocin ointment or fusidic acid** as a topical antibiotic
- If systemic symptoms present, use **cloxacillin** or a **beta-lactamase resistant antibiotic** e.g. cephalexin
- If impetigo is due to **MRSA**, use **clindamycin**

## Cellulitis

Cellulitis is a **serious bacterial skin infection that affects the deeper layers of the skin and underlying tissues**.

It occurs when bacteria normally found on the skin surface **breach the protective barrier**.

It appears as a **painful, red, swollen, and warm area that can spread quickly**.

Primarily caused by *Staphylococcus* (staph) or *Streptococcus* (strep) bacteria entering through a break in the skin.

## Predisposing and risk factors

1. **Skin injuries** – cuts, surgical wounds, insect bites, tattoos, or piercings
2. **Pre-existing skin conditions** – eczema, athlete's foot, or psoriasis
3. **Weakened immune system** – diabetes, HIV, or use of immunosuppressant medications
4. **Poor circulation** – swelling (lymphoedema) or obesity

## Clinical features

1. Redness and swelling of the skin
2. Pain or tenderness to the touch
3. A feeling of warmth in the affected area
4. Blisters, fever, or swollen lymph nodes in severe cases

## Treatment

- **Oral & IV antibiotics** – first-line treatments should target common bacteria like *Streptococcus* and *Staphylococcus*
- **Wound care & drainage** – any open wounds or cuts should be kept clean and covered to help speed up healing
- **Elevation & rest** – resting the affected area and elevating it above the level of the heart reduces swelling and improves blood flow
- **Pain management** – analgesics such as ibuprofen or paracetamol, to reduce pain and fever

## Complications

- Bacteraemia
- Endocarditis
- Osteomyelitis
- Suppurative arthritis
- Thrombophlebitis

## Staphylococcal scalded skin syndrome (SSSS) / Ritter's disease

Caused by **exfoliative toxins A and B** produced by *Staphylococcus aureus* belonging to **phage group B**.

**Scarlatiniform erythema** develops diffusely, with a predilection for **flexural and periorificial areas**.

There is development of **extensive superficial sterile flaccid blisters or bullae**, which become derroofed, revealing **large areas of raw red moist skin similar to a hot water burn**.

**Fever, rhinitis, conjunctivitis, irritability, malaise, poor feeding and exquisite tenderness** may occur prior to the onset of the rash.

**Circumpolar erythema, radial crusting and fissuring around the eyes, mouth, and nose** are characteristically prominent. The child may appear **toxic**.

**Nikolsky's sign** – separation of areas of epidermis in response to shearing pressure on the skin – is **positive**.

### Treatment

- Antibiotics, e.g. **flucloxacillin, erythromycin**
- Analgesics
- Application of an **emollient**; clean with **isotonic saline**
- **Correct fluid and electrolyte balance**